

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2024-2025
FULL MARKS: 150

CSE 4835: Pattern Recognition

See Page 4 for appendix. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) State whether each statement is **True** or **False** and justify your answer briefly. 5 × 3
(CO1)
(PO1)
- If Model A has a lower test loss on a dataset than Model B, then Model A must have a higher accuracy on the test dataset than Model B.
 - Consider the loss function

$$L(w) = \frac{1}{N} \sum_{i=1}^N L_i(f(x_i, w), y_i) + \lambda R(w), \quad R(w) = \|w\|_2^2,$$
 where L_i is a standard training loss. If $\lambda = -1$, then for every w there exists another set of weights w' with lower loss and accuracy less than or equal to that of w .
 - ReLU (Rectified Linear Unit) activation functions can suffer from the problem of dead neurons.
 - You know that you have a good training setup if there is no gap between the training accuracy and validation accuracy.
 - Consider the two models below. Both take a $4 \times 4 \times 3$ image as input and produce scores for 10 classes. Model 1 has more learnable parameters than Model 2.

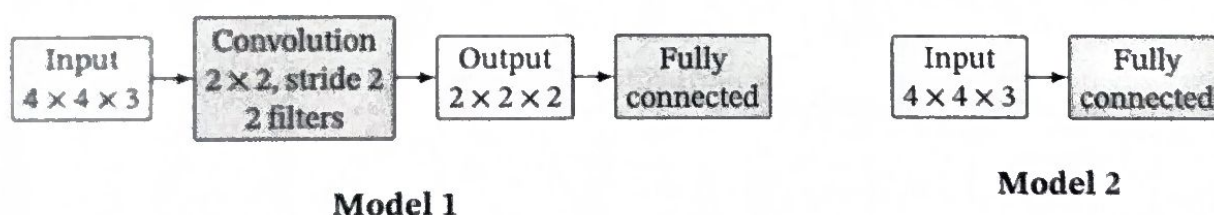


Figure 1: Architectures used in Statement (v) of Question 1.a.

- b) Suppose your Pattern Recognition course has 60 students, and you collect handwritten character samples from them. Thirty students write in cursive, while the other thirty write in block letters.
- If you had to build a character-recognition system from these samples, would you prefer a k -Nearest Neighbour classifier or a linear classifier? Explain your choice with reference to the two writing styles. 5
(CO3)
(PO3)
 - If you knew which students wrote in cursive and which wrote in block letters, how would you change the training or prediction strategy for each of the two classifiers? 5
(CO3)
(PO3)
2. a) Consider three fully connected linear layers stacked one after another.
- Does this have greater representational power than a single linear classifier? Justify your answer. 5
(CO1)
(PO1)
 - How does inserting ReLU activations between the layers change your answer? Explain using the concept of space warping. 5
(CO1)
(PO1)

- b) The figure below shows the training loss curves for the same network and optimizer, trained for the same number of epochs using five different constant learning rates.

5 + 5
(CO2)
(PO2)

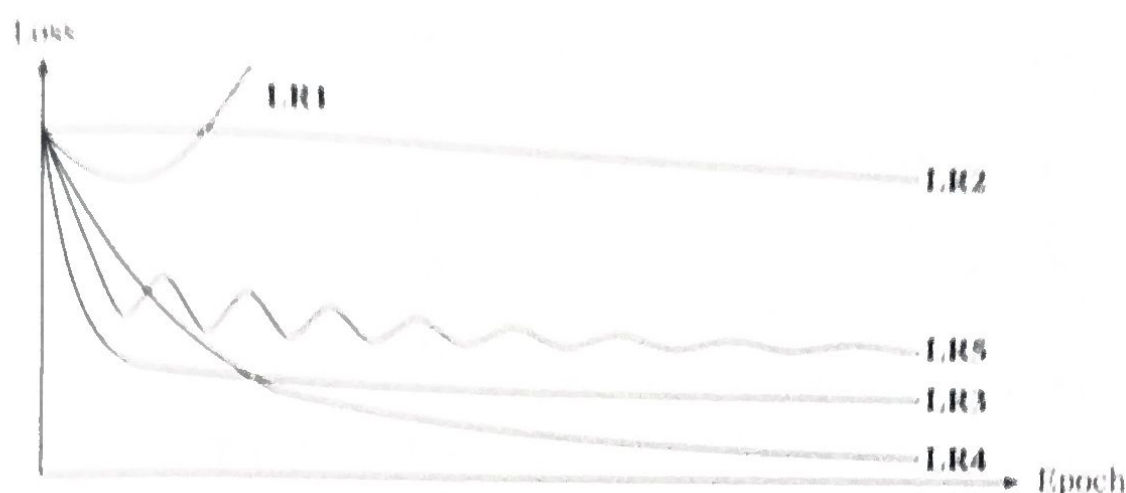


Figure 2: Training loss for five constant learning rates.

Arrange LR1–LR5 in descending order of learning rate. Justify your ordering by explaining what the qualitative behaviour of each curve indicates about its step size.

- c) During training, the gradient of parameter w_1 repeatedly changes sign and has a large magnitude, while the gradient of parameter w_2 is smaller but consistently has the same sign. Explain how SGD with momentum and Adam would treat these two parameters, and identify which behaviour is useful in each direction.
3. a) Due to a printer fault, most images of the digit 7 in a dataset contain an ink blot in the bottom-right corner (as seen in Images 1, 2 & 4 in Figure 3); no image of any other digit contains this blot. A convolutional neural network was trained on this dataset and then evaluated on the four images in Figure 3. Beside each image, its Grad-CAM explanation is shown: darker regions indicate greater importance to the prediction.

5
(CO2)
(PO2)

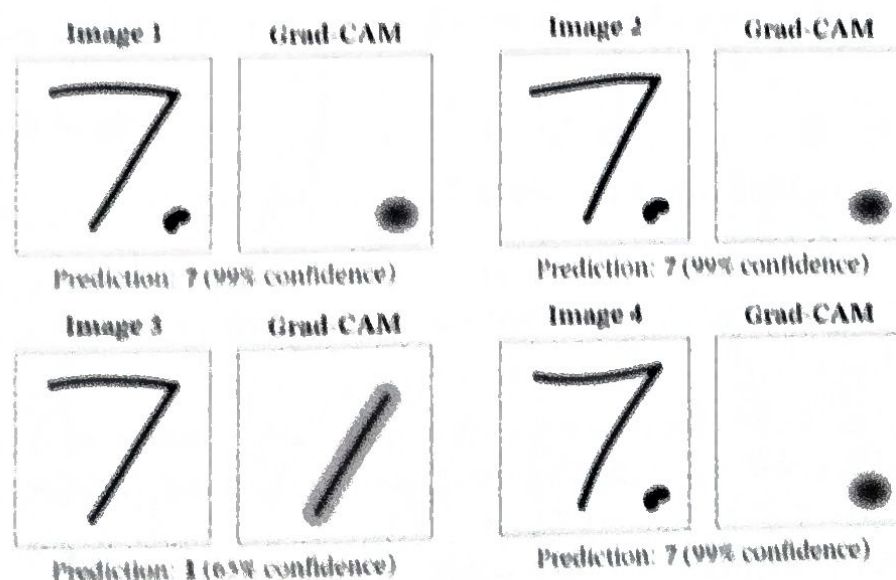


Figure 3: Four images of the digit 7 and their Grad-CAM explanations.

- Explain the purpose of Grad-CAM and briefly describe how it produces a class-specific visual explanation.
- Based on the predictions and Grad-CAM explanations, explain why Image 3 is misclassified.
- Propose a modification to the training strategy that would fix this failure mode.

5
(CO1)
(PO1)
5
(CO2)
(PO2)
5
(CO3)
(PO3)

- b) A traffic-sign classifier is trained on a small dataset containing left-turn, right-turn, speed-limit, and traffic-light classes. A student proposes horizontal flipping, random cropping, and strong colour jitter as augmentations to use during training.

i. Evaluate the suitability of each proposed augmentation for this task.

6

(CO2)

(PO2)

ii. After applying suitable augmentations, training accuracy decreases while validation accuracy improves. Explain why this is not necessarily a problem.

4

(CO2)

(PO2)

4. a) One of the primary design philosophies behind GoogLeNet was compute efficiency. It used an aggressive stem near the input and global average pooling near the classifier. Describe these two design choices used at these locations and explain how each improves efficiency. Discuss any information or representational trade-offs.

5 + 5

(CO1)

(PO1)

- b) The Inception module, as shown in Figure 4 uses parallel branches and 1×1 bottlenecks before its 3×3 and 5×5 convolutions.

5 + 5

(CO1)

(PO1)

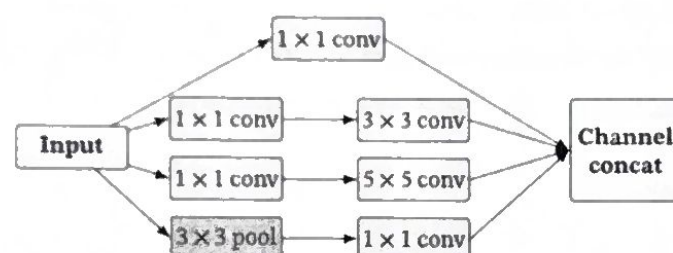


Figure 4: Inception module used in GoogLeNet.

Explain the purposes of the following: (i). the parallel branches and (ii). the 1×1 convolutions before the larger convolutions.

- c) ShuffleNet uses grouped convolutions and channel shuffle. Explain why fixed groups restrict information flow, and how channel shuffle restores communication while retaining the efficiency of grouped convolutions.

7

(CO1)

(PO1)

- d) Figure 5 compares a standard 3×3 convolution with a depthwise separable block.

5 + 3

(CO1)

(PO1)

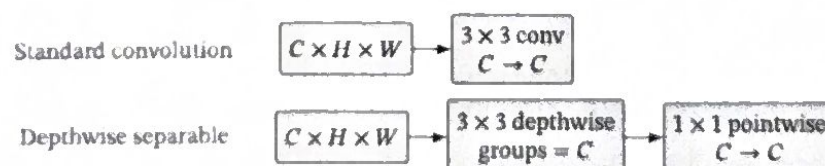


Figure 5: Standard convolution vs. depthwise separable convolution.

Explain: (i). roles of the depthwise and pointwise stages, why both are needed, and (ii). why the separable block is computationally cheaper than the standard convolution.

5. a) A CNN pretrained on ImageNet (a large dataset containing classes such as dogs, birds, and vehicles) needs to be adapted via transfer learning to each of the following image classification datasets:

- Dataset A contains 100 labelled images from three animal classes: koala, otter, and platypus.
- Dataset B contains 5,000 labelled chest X-rays from two classes: normal and pneumonia.

- i. Suppose a student trains the new classifier but freezes the later layers while fine-tuning the earlier layers. Explain why reversing the usual strategy in this way is generally ineffective and what problems it could cause.

7

(CO2)

(PO2)

- ii. Compare the extent to which you would freeze or fine-tune the model across the two datasets. Identify which dataset would require more fine-tuned layers,

7

(CO2)

(PO2)

3 + 3
(CO1)
(PO1)

- b) Consider a deep ReLU network. Explain what happens to its activations and gradients as they pass through many layers when the initial weights are: (i). too small; and (ii). too large.
6. a) The two convolutional networks below take a 32×32 RGB image as input and produce scores for 10 classes.

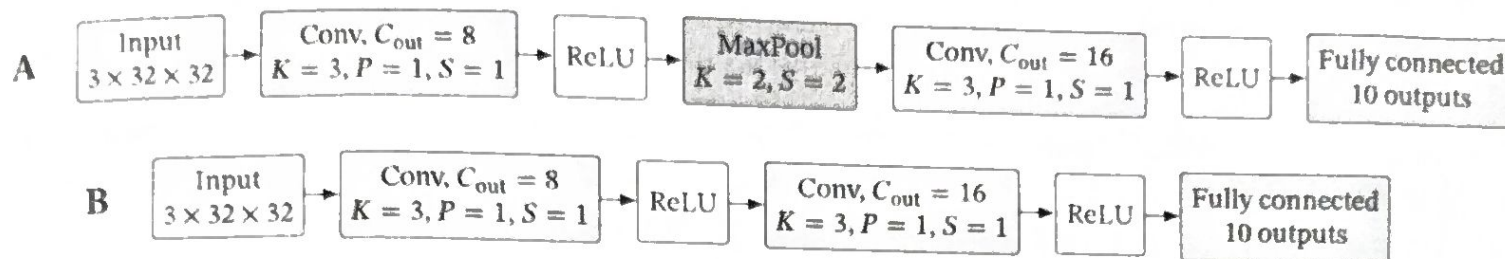


Figure 6: Architectures of Network A and Network B.

- i. For each network, determine:
- the output size after each layer; and
 - the number of learnable parameters at each layer and in the complete network.
- ii. Compare Networks A and B. Explain the advantages and disadvantages of including the max-pooling layer.

14
(CO1)
(PO1)
6
(CO2)
(PO2)

Appendix for CSE 4835

Loss functions and regularisation

Scores $\mathbf{s} = W\mathbf{x} + \mathbf{b}$; correct class y_i .
 $L_i^{\text{SVM}} = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + \Delta)$, $\Delta = 1$,

$$L_i^{\text{CE}} = -\log \frac{e^{s_{y_i}}}{\sum_j e^{s_j}},$$

$$L = \frac{1}{N} \sum_i L_i + \lambda R(W),$$

Optimisation updates

SGD: $w_{i+1} = w_i - \alpha g_i$,

Momentum: $v_i = \rho v_{i-1} + g_i$, $w_{i+1} = w_i - \alpha v_i$,

AdaGrad: $G_i = G_{i-1} + g_i^2$, $w_{i+1} = w_i - \alpha \frac{g_i}{\sqrt{G_i + \epsilon}}$,

RMSProp: $G_i = \beta G_{i-1} + (1 - \beta) g_i^2$,

$$w_{i+1} = w_i - \alpha \frac{g_i}{\sqrt{G_i + \epsilon}},$$

Adam: $m_i = \beta_1 m_{i-1} + (1 - \beta_1) g_i$, $v_i = \beta_2 v_{i-1} + (1 - \beta_2) g_i^2$,

$$\hat{m}_i = \frac{m_i}{1 - \beta_1^i}, \quad \hat{v}_i = \frac{v_i}{1 - \beta_2^i},$$

$$w_{i+1} = w_i - \alpha \frac{\hat{m}_i}{\sqrt{\hat{v}_i + \epsilon}}.$$

Convolutional networks

Input $C_{\text{in}} \times H \times W$; kernel K , padding P , stride S .
 $H' = \frac{H - K + 2P}{S} + 1$, $W' = \frac{W - K + 2P}{S} + 1$,

$$P = \frac{K - 1}{2} \quad (\text{same}, S = 1).$$

$$\# \text{parameters} = C_{\text{out}}(C_{\text{in}}K^2 + 1),$$

$$\# \text{multiply-adds} = C_{\text{out}}H'W'C_{\text{in}}K^2,$$

$$\text{RF}(L \text{ stride-1 layers}) = 1 + L(K - 1).$$

Batch normalisation

$$\mu_j = \frac{1}{N} \sum_i x_{i,j}, \quad \sigma_j^2 = \frac{1}{N} \sum_i (x_{i,j} - \mu_j)^2,$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}, \quad y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j.$$

Normalisation axes for $N \times C \times H \times W$

BN	N, H, W (per channel)
LN	C, H, W (per example)
IN	H, W (per example/channel)
GN	$C/G, H, W$ (per example/group)

Activations and initialisation

$\text{ReLU}(x) = \max(0, x)$, $\text{LReLU}(x) = \max(x, \alpha x)$,

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad \sigma'(x) = \sigma(x)(1 - \sigma(x)),$$

$$\text{std}(W)_{\text{Xavier}} = \frac{1}{\sqrt{D_{\text{in}}}}, \quad \text{std}(W)_{\text{Kaiming}} = \sqrt{\frac{2}{D_{\text{in}}}}.$$

Learning-rate schedules

For $u = t/T \in [0, 1]$:

$$\alpha_i^{\text{cosine}} = \frac{\alpha_0}{2}(1 + \cos \pi u), \quad \alpha_i^{\text{linear}} = \alpha_0(1 - u),$$

$$\alpha_i^{\text{invsqrt}} = \alpha_0/\sqrt{t}, \quad \alpha_i^{\text{step}} = \alpha_0 \gamma^k.$$

Visualising models

For class score s_c , feature maps $A^k \in \mathbb{R}^{H \times W}$, and $Z = HW$:

$$\text{CAM: } M_c(i, j) = \sum_k w_{ck} A_{ij}^k.$$

$$\text{Grad-CAM: } \alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial s_c}{\partial A_{ij}^k},$$

$$L_{\text{GradCAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right).$$